

Seismology Problem Set 2

David Al-Attar

Michaelmas Term, 2026

1. Consider the regularised least squares solution to a linear(ised) inverse problem obtained by minimising the function

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m} - \mathbf{d}) + \frac{\lambda}{2} \mathbf{m}^T \mathbf{B}\mathbf{m}.$$

Suppose that the data take the form $\mathbf{d} = \mathbf{A}\mathbf{m}_{\text{in}}$ with $\mathbf{m}_{\text{in}}$ a specified input model. Show that the resulting least squares solution, $\mathbf{m}_{\text{out}}$, takes the form

$$\mathbf{m}_{\text{out}} = \mathbf{R}\mathbf{m}_{\text{in}},$$

where $\mathbf{R}$ is a matrix to be determined. Discuss the significance of this so-called resolution matrix, explaining why, in particular, it would be desirable for its value to be close to the identity matrix.

2. Consider a linear inverse problem of the standard form

$$\mathbf{d} = \mathbf{A}\mathbf{m} + \mathbf{e},$$

with $\mathbf{d}$ an n -dimensional data vector, $\mathbf{m}$ an m -dimensional model vector, $\mathbf{A}$ an $n \times m$ matrix, and $\mathbf{e}$ an n -dimensional vector of random errors. We assume $m > n$, which is to say that there are more unknown model parameters than available data. We further suppose that the data have been processed such that the components of the random errors are uncorrelated and have unit standard deviations (this corresponds to the data covariance, $\mathbf{C}$, being the $n \times n$ identity matrix).

- (a) A model $\mathbf{m}$ is said to be compatible with the data if

$$\|\mathbf{A}\mathbf{m} - \mathbf{d}\|^2 \leq \chi_c^2,$$

where χ_c^2 is a critical value for the so-called chi-squared statistic. What type of geometric region of the model space is defined by this inequality?

- (b) Supposing that $\|\mathbf{d}\|^2 > \chi_c^2$, the region of the model space defined by the inequality in (a) does not contain the origin. In such cases, the closest point within this region to the origin must lie on the boundary set defined by

$$\|\mathbf{A}\mathbf{m} - \mathbf{d}\|^2 = \chi_c^2.$$

Using the method of Lagrange multipliers, we can determine this *minimum norm solution* by considering the Lagrangian

$$\mathcal{L}(\mathbf{m}, \mu) = \|\mathbf{m}\|^2 + \mu(\|\mathbf{A}\mathbf{m} - \mathbf{d}\|^2 - \chi_c^2),$$

where μ is a Lagrange multiplier. Derive the equations for $\mathbf{m}$ and μ that must hold at the minimum norm solution. Discuss the relationship between this approach and regularised least squares.

3. Consider the deformation of an isotropic elastic body due to an applied surface load, σ . The relevant linearised equations of motion are

$$-\frac{\partial T_{ij}}{\partial x_j} = 0,$$

$$T_{ij} = \lambda \frac{\partial u_k}{\partial x_k} \delta_{ij} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right),$$

within the reference body M , and subject to boundary conditions

$$T_{ij}\hat{n}_j = \sigma g_i$$

on its boundary, ∂M . Here g_i is the acceleration due to gravity prior to the deformation. Note that self-gravitation is being ignored within these equations.

Suppose that observations, $\mathbf{u}_i^{\text{obs}}$, of the displacement vector have been made at surface locations $\mathbf{x}_i$ for $i = 1, \dots, m$. An inverse problem could then be formulated to estimate the Lamé parameters, λ and μ , as well as the load, σ . To this end, we can seek to minimise the misfit

$$J = \frac{1}{2} \sum_{i=1}^m \frac{1}{\varepsilon_i^2} \|\mathbf{u}(\mathbf{x}_i) - \mathbf{u}_i^{\text{obs}}\|^2,$$

where ε_i is the standard deviation for the i th observation. Adapt the approach within Lecture 19 to derive sensitivity kernels for J with respect to λ , μ , and σ , showing how they can be calculated through one solution of the forward problem and one of an appropriate adjoint problem.

4. Recall that the Lagrangian for a self-gravitating elastic body is given by

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2} \rho(\mathbf{x}) \|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2} \rho(\mathbf{x}) \zeta(\mathbf{x}, t),$$

where ζ is the referential gravitational potential

$$\zeta(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)][\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]\}^{\frac{1}{2}}} d^3\mathbf{x}'.$$

By calculating the appropriate functional derivative, show that

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho \gamma_i,$$

where γ_i is the referential gravitational acceleration.

5. Consider a planet at equilibrium occupying the volume M and with density ρ . Its gravitational potential, ϕ , is a solution of Poisson's equation

$$\nabla^2 \phi = 4\pi G \rho,$$

where it is understood that ρ vanishes outside of M . Across the surface ∂M we then have continuity conditions on ϕ and its normal derivative, and finally require that ϕ tend to zero at infinity. Suppose that the planet is only slightly aspherical, with boundary

$$r = b + sh_1(\theta, \varphi) + \dots,$$

in spherical polar co-ordinates, and its density decomposed as

$$\rho = \rho_0 + s\rho_1 + \dots,$$

with ρ_0 spherically symmetric. Here s is a perturbation parameter, and the dots denote higher-order terms that are to be neglected. Similarly writing the potential in the form $\phi = \phi_0 + s\phi_1 + \dots$, show that across the reference surface $r = b$ we require continuity of both ϕ_1 and

$$\frac{\partial \phi_1}{\partial r} + 4\pi G \rho_0 h_1.$$

6. Using results from Lecture 22 along with residue calculus (which you don't need to know for this course), it can be shown that the time-domain solution of the elastodynamic equations in a non-rotating earth model can be expressed as the following eigenfunction expansion

$$\mathbf{u}(\mathbf{x}, t) = \sum_{km} \left\{ \int_0^t \frac{\sin[\omega_k(t-t')]}{\omega_k} \int_M \bar{S}_{ij}(\mathbf{x}', t') \frac{\partial |km\rangle^*}{\partial x'_j}(\mathbf{x}') d^3\mathbf{x}' dt' \right\} |km\rangle(\mathbf{x}).$$

Simplify this expression in the case of a point source for which

$$\bar{S}_{ij}(\mathbf{x}, t) = M_{ij} \delta(\mathbf{x} - \mathbf{x}_s) H(t - t_s),$$

with M_{ij} known as the moment tensor, $\mathbf{x}_s$ the source location, and t_s the source time. Describe briefly and qualitatively how an inverse problem might be formulated to estimate the ten source parameters ($M_{ij}, \mathbf{x}_s, t_s$) from recorded seismograms.

7. Consider the eigenvalue problem for a rotating earth model

$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + i\omega \langle \mathbf{w} | W | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0,$$

where $\mathbf{s}$ is the eigenfunction, ω the eigenfrequency, and $\mathbf{w}$ an arbitrary test function. Assuming the eigenfunction normalisation condition

$$\langle \mathbf{s} | P | \mathbf{s} \rangle = 1,$$

show that the eigenfrequency can be written

$$\omega = \frac{1}{2} i \langle \mathbf{s} | W | \mathbf{s} \rangle \pm \sqrt{\langle \mathbf{s} | H | \mathbf{s} \rangle - \frac{1}{4} \langle \mathbf{s} | W | \mathbf{s} \rangle^2}.$$

Use this result to discuss qualitatively the stability of a rotating earth model, including, in particular, the role of the Coriolis and centrifugal forces.